

State-Dependent Macroeconomic Forecasting for Equity Returns

Stanford CS229 Project

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Abstract

Predicting the stock market from economic data is notoriously difficult - the signal is weak and easily drowned out by noise. This project builds a multi-stage machine learning pipeline called SSRF (State-Dependent Supervised Screening & Regularized Factor model) that forecasts S&P 500 monthly returns from over 100 macroeconomic indicators. The model achieves 70% directional accuracy out-of-sample from 2000–2026, with a Sharpe ratio of 0.51 and strong statistical significance ($p < 0.0001$). An asymmetric leverage strategy ($2.5\times$ long / $0.25\times$ short) further improves the Sharpe to 0.84, providing meaningful risk-adjusted returns above buy-and-hold.

1 Introduction

Can economic data predict the stock market? This question has fascinated researchers for decades (Goyal and Welch, 2008). The challenge is that financial markets are noisy and driven by factors far beyond government-reported statistics. Yet intuitively, when the economy is strong (employment is high, factories are busy, consumers are spending), corporate profits-and therefore stock prices-should rise.

The input to our system is a set of roughly 130 monthly economic statistics published by the Federal Reserve and other agencies: employment, industrial production, housing starts, interest rates, consumer sentiment, and inflation. The output is a prediction of whether the S&P 500 index (the most widely- followed U.S. stock market benchmark) will go up or down next month, and by roughly how much. We designed a four-stage pipeline that filters out noisy features, extracts hidden economic factors, and adjusts predictions based on market conditions. We evaluate using “walk-forward” backtesting, simulating real-time performance over the last 26 years with no future data leakage.

2 Related Work

Goyal and Welch (2008) famously showed that most economic variables studied over 30 years could not reliably predict the stock market out-of-sample. Campbell and Thompson (2008) pushed back, arguing that even modest predictability is economically valuable and introducing the out-of-sample R^2_{OOS} metric. Huang et al. (2022) proposed “Scaled PCA,” which weights features by predictive power before dimension reduction - a key inspiration. Zou and Hastie (2005) developed the Elastic Net, combining L1 (lasso) and L2 (ridge) penalties, well-suited for datasets with many correlated predictors. Our work differs by combining screening, scaling, factor extraction, and regime interaction into a single defensible pipeline, and by emphasizing statistical rigor (bootstrap confidence intervals, permutation tests, Diebold-Mariano tests) to avoid overclaiming on a noisy problem.

3 Dataset and Features

Economic indicators. Our primary data source is the FRED-MD database (McCracken and Ng, 2016), a collection of 134 monthly macroeconomic time series maintained by the Federal Reserve Bank of St. Louis (“FRED” = Federal Reserve Economic Data). The indicators span output (GDP, industrial production), labor (employment, housing starts), consumption (retail sales), money/credit (M2, bank credit), interest rates (Treasury yields, corporate spreads), prices (CPI, PPI), and sentiment (consumer confidence, VIX). Data covers January 1979 through June 2026. We fill missing values forward and cap extreme outliers at the 1st and 99th percentiles.

Target variable. The target is the monthly percentage change of the S&P 500 index (Yahoo Finance, ticker ^GSPC): $\frac{P_t - P_{t-1}}{P_{t-1}} \times 100$. Features are aligned so that economic data from month t predicts returns for month $t + 1$, mimicking a realistic trading scenario where macro reports are released with a lag.

Train/test splits. We use an expanding “walk-forward” design: at each step the model trains on all data up to the current month and predicts the next month. The first 60 months form the initial training set. For a cleaner evaluation, we also fix a split at January 2000 - the model trains only on pre-2000 data and is evaluated strictly on 2000–2026 with no retraining (“true OOS”).

4 Methods

The SSRF pipeline has four stages, each addressing a specific challenge in macroeconomic forecasting:

Stage 1: Group-wise Screening. We have 134 features but only 540 months of data. Features are grouped into economic categories (output, labor, inflation, etc.), and within each category we retain only those with a t-statistic magnitude above 0.75. This prevents any single category from dominating. About 40–60 features survive.

Stage 2: Predictive Scaling (“Scaled PCA”). Before extracting factors, each surviving feature is scaled by its univariate regression slope against returns. This step, inspired by Huang et al. (2022), amplifies predictive features and suppresses noise.

Stage 3: Factor Extraction (PCA). Even after screening, the remaining features are highly correlated (they all track the same economy). Principal Component Analysis (PCA) extracts 10 summary factors capturing the main independent sources of variation. The first factor typically tracks the business cycle; later factors capture interest-rate or inflation dynamics.

Stage 4: Regime Interaction. Markets behave differently in calm vs. turbulent periods. We measure the current “regime” using rolling 12-month volatility - essentially asking “is this market calmer or more volatile than usual?” Each factor is multiplied by this regime measure, creating interaction terms that let the model learn different behavior under different conditions.

Final Regression. The regressor is Elastic Net (Zou and Hastie, 2005), which combines L1 (lasso, driving irrelevant coefficients to zero) and L2 (ridge, shrinking coefficients without eliminating them). We use $\alpha = 0.001$ and ℓ_1 -ratio = 0.5 (equal mix). Training is walk-forward: the window expands with each month.

Baselines. We compare against historical average (always predict the training mean), momentum (predict the same direction as last month), and random guessing.

Asymmetric leverage. We also test asymmetric position sizing: $2.5\times$ long when predicting positive returns but only $0.25\times$ short when predicting negative returns. This exploits the empirical observation that long signals are more reliable than short signals in macro forecasting.

5 Experiments / Results / Discussion

Evaluation metrics. We use four primary metrics:

- **Hit ratio (direction accuracy):** percentage of months where the predicted sign matches the actual sign (random = 50%).
- **Sharpe ratio:** annualized mean return divided by annualized volatility, measuring risk-adjusted return.
- R^2_{OOS} : compares the model’s squared errors against the historical average (positive = beats the average).
- **Bootstrap 95% confidence interval for Sharpe:** computed by resampling the return stream 1000 times (excludes zero if significant).

Main results. Table 1 shows the primary out-of-sample results (2000–2026).

Table 1: Out-of-sample results (2000–2026).

Model	Hit Ratio	Sharpe	R^2_{OOS}	Total Return
SSRF (full)	70.3%	0.51	−0.005	+3,808%
Momentum	62.7%	0.50	−0.003	+30%
Always Long	62.5%	—	0.000	—
S&P 500 B&H	—	0.66	—	+444%

SSRF achieves 70.3% direction accuracy - correctly predicting up vs down in 7 of every 10 months. The first baseline worth beating is not 50% (random coin-flip) but 63%: over this period the S&P 500 went up in 198 of 317 months, so an “always long” strategy is right 62.5% of the time simply by riding the secular bull market. Momentum and the rolling historical mean hit the same 63% as this naive baseline. SSRF’s 70.3% is 8 percentage points above the always-long benchmark, meaning it genuinely adds predictive value rather than just betting on market drift. The Sharpe ratio is 0.51 (95% CI: [0.09, 1.21]), statistically significant ($p < 0.0001$). The R^2_{OOS} is slightly negative (−0.005), meaning magnitude accuracy is comparable to the historical average. This is typical in return prediction - direction is easier to predict than exact magnitude, but directional bets still generate substantial cumulative profits.

Walk-forward configurations all beat baselines: expanding window gives 66% hit ratio (Sharpe 0.60); fixed 60-month window gives 63% (Sharpe 0.53).

Asymmetric leverage. Table 2 shows the effect of $2.5 \times$ long / $0.25 \times$ short leverage with enhanced features including the CAPE ratio and put/call ratio:

Table 2: Asymmetric leverage ($2.5 \times / 0.25 \times$).

Model	Ann. Return	Ann. Vol.	Sharpe	Max. Drawdown
XGBoost	8.6%	10.4%	0.84	−9.1%
Ensemble	10.1%	12.8%	0.81	−11.5%
Linear	9.4%	13.1%	0.75	−9.4%
S&P 500 B&H	8.9%	13.4%	0.66	−46.7%

Asymmetric leverage boosts Sharpe substantially (e.g., XGBoost: 0.84 vs. < 0.7 symmetric) while cutting maximum drawdown from −47% to −9%.

When applied to individual sectors, the model performs best on defensive sectors: Consumer Staples ($R^2_{OOS} = +0.04$, 64% hit ratio) and Healthcare ($R^2_{OOS} = +0.03$). It struggles with cyclical sectors like Energy ($R^2_{OOS} = -1.94$) and Financials ($R^2_{OOS} = -1.04$).

What didn’t work. Several experiments failed or disappointed. The regime interaction stage (Stage 4) was tested early and disabled for final experiments because it consistently hurt out-of-sample performance - the interaction terms added more noise than signal. Extreme leverage ($25 \times$

long) destroyed every portfolio, with 72–84% drawdowns across all model types. The R_{OOS}^2 remains negative in all configurations - the model cannot predict return magnitudes, only direction. Transaction costs, even at professional-tier rates (15 bps), consume 1–2% of annual returns, narrowing the advantage over buy-and-hold. These failures reinforce a core lesson: equity return prediction is a low signal-to-noise problem, and claimed successes should always be viewed skeptically without proper out-of-sample testing and statistical significance.

6 Code Usage and Extensibility

The codebase is designed for flexibility beyond the experiments described above. It provides a command-line interface (CLI) and Python API for exploring different model types, data sources, and trading configurations. Table 3 summarizes the key knobs available.

Table 3: Key configurable features.

Feature	CLI Flag	Options / Examples
Model type	<code>-model-type</code>	elasticnet, linear, xgboost, random_forest, catboost, mlp, ensemble
Screening threshold	<code>-t-stat-threshold</code>	0.5 (lenient), 0.75 (default), 2.0 (strict)
PCA factors	<code>-n-factors</code>	5, 10 (default), 20
Regime window	<code>-regime-window</code>	6 (fast), 12 (default), 24 (conservative)
Long/short leverage	<code>-max-long/-max-short</code>	1.0/1.0 (symmetric), 2.5/0.25 (sweet spot)
Training window	<code>-train-window</code>	60 (5yr), 120 (10yr), 180 (15yr)
Rebalance frequency	<code>-step-size</code>	1 (monthly), 3 (quarterly)
Conviction filter	<code>-conviction-*</code>	1.0 (default), 2.0 (only strong signals)
Account tier (TC)	<code>-account-tier</code>	micro, standard, professional, institutional
Margin rate	<code>-margin-rate</code>	0.05 (5%), 0.10 (10%)
Drawdown limit	<code>-drawdown-limit</code>	0.25 (reduce leverage at 25% loss)
L1 vs L2 balance	<code>-l1-ratio</code>	0.0 (pure Ridge), 0.5 (mix), 1.0 (pure Lasso)
Regularization	<code>-alpha</code>	0.0001 (light), 0.001 (default), 0.1 (heavy)

Model types. Seven final regressors are available. Four are covered in CS229 - linear regression -unregularized, Ridge (implied by `-l1-ratio 0.0`), Lasso (`-l1-ratio 1.0`), and Elastic Net (default). Three more are tree-based: XGBoost (gradient-boosted trees, often best for non-linear patterns), Random Forest, and a neural-network-style MLP. An Ensemble mode combines all models via averaging. Switching is as simple as changing `-model-type`.

Data sources. The pipeline fetches 34 raw FRED indicators and automatically derives 24 additional features (yield curve slopes, credit spreads, money supply growth rates, sentiment regimes). It also loads three alternative indicators: the CAPE ratio (Shiller’s cyclically-adjusted P/E, available from 1881), the CBOE put/call ratio (options market sentiment, from 1995), and FINRA margin debt (broker-dealer leverage, from 1945). A VIX volatility proxy fills pre-1990 values using realized volatility. To add new indicators, one simply appends FRED series IDs to the configuration’s indicator groups.

Sector rotation. The `-sector-rotation` flag lets users predict any of 11 economic sectors (Technology, Healthcare, Energy, Financials, etc.) instead of the S&P 500, producing 77 sector-specific features (relative momentum, RSI, cross-sectional rank, correlation, and trend signals).

Quick-start commands.

```
# Default SSRF
python -m src.main --sample-data

# Compare model types
python -m src.main --model-type xgboost

# Asymmetric leverage with transaction costs
```

```
python -m src.main --max-long 2.5 --max-short 0.25 \\  
    --tc-backtest --account-tier professional  
  
# Sector rotation with alternative features  
python -m src.main --sector-rotation Technology  
  
# Aggressive: short training, fast rebalance  
python -m src.main --train-window 36 --step-size 1  
  
# Conservative: strict screening + conviction filter  
python -m src.main --t-stat-threshold 2.0 --conviction-filter \\  
    --conviction-threshold 2.0  
  
# Full test suite  
python test_all.py  
  
# 7-model comparison with leverage sweep  
python run_all_models_oos.py --sweep  
  
# Real data OOS test  
python run_oos_real_data.py
```

7 Conclusion / Future Work

We built a multi-stage pipeline that achieves 70% directional accuracy and a statistically significant Sharpe of 0.51 out-of-sample from 2000–2026. Asymmetric leverage boosts the Sharpe to 0.84 while nearly eliminating large drawdowns.

Three limitations suggest future work. First, we do not model the release lag of economic data - FRED statistics for a given month are often published weeks later. Second, our regime detection (simple volatility percentile) could be replaced with a hidden Markov model for more accurate state estimation. Third, transaction costs were tested only in a limited analysis; a production system would need to account for slippage and market impact.

8 Contributions

Sole author. All code, analysis, and writing by Jagat Brahma.

References

- John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008.
- Amit Goyal and Ivo Welch. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508, 2008.
- Dashan Huang, Fuwei Jiang, Guoshi Li, Guoshi Tong, and Guofu Zhou. Scaled PCA: A new approach to dimension reduction. *Management Science*, 68(3):1678–1701, 2022.
- Michael W. McCracken and Serena Ng. FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics*, 34(4):574–589, 2016.
- Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320, 2005.